

Inflammatory Jaw Lesions - Histology

OMFS 661: Diagnostic Oral Radiology and Pathology

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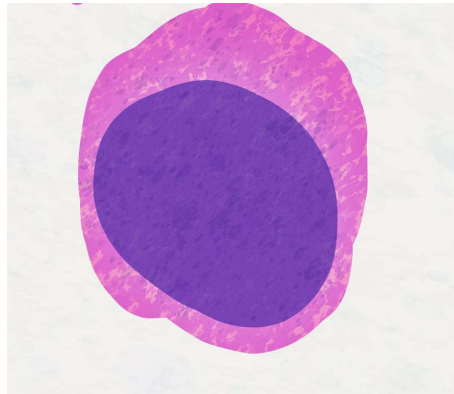
Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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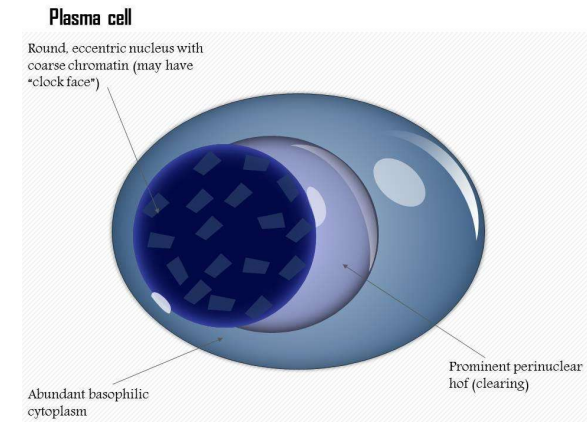
Types of Inflammatory Cells



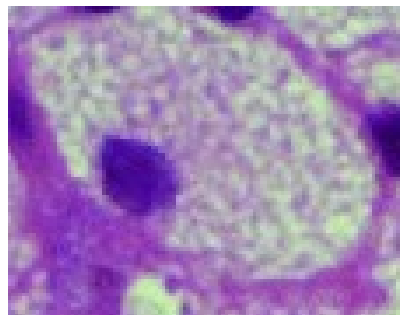
Neutrophil



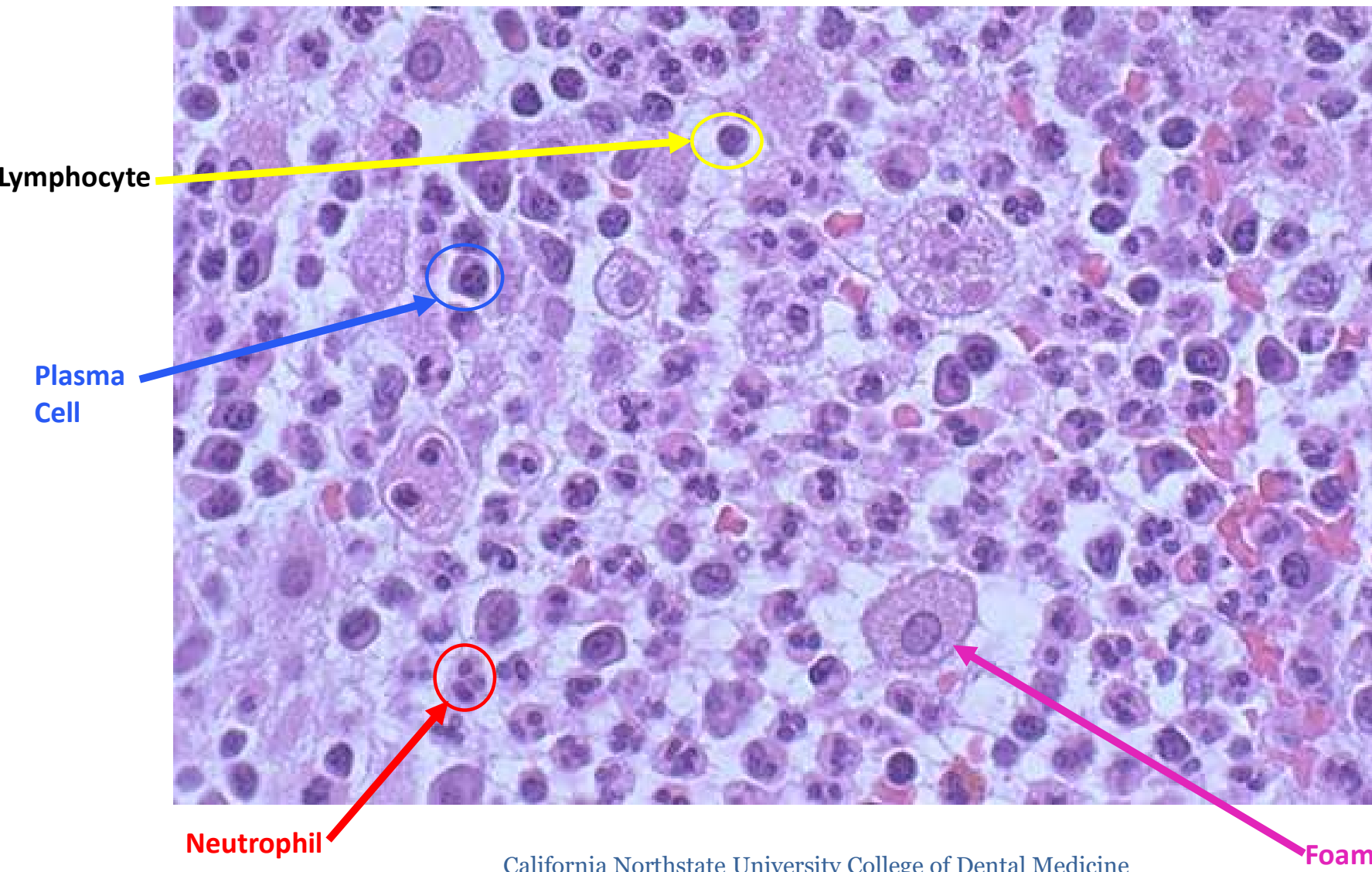
Lymphocyte



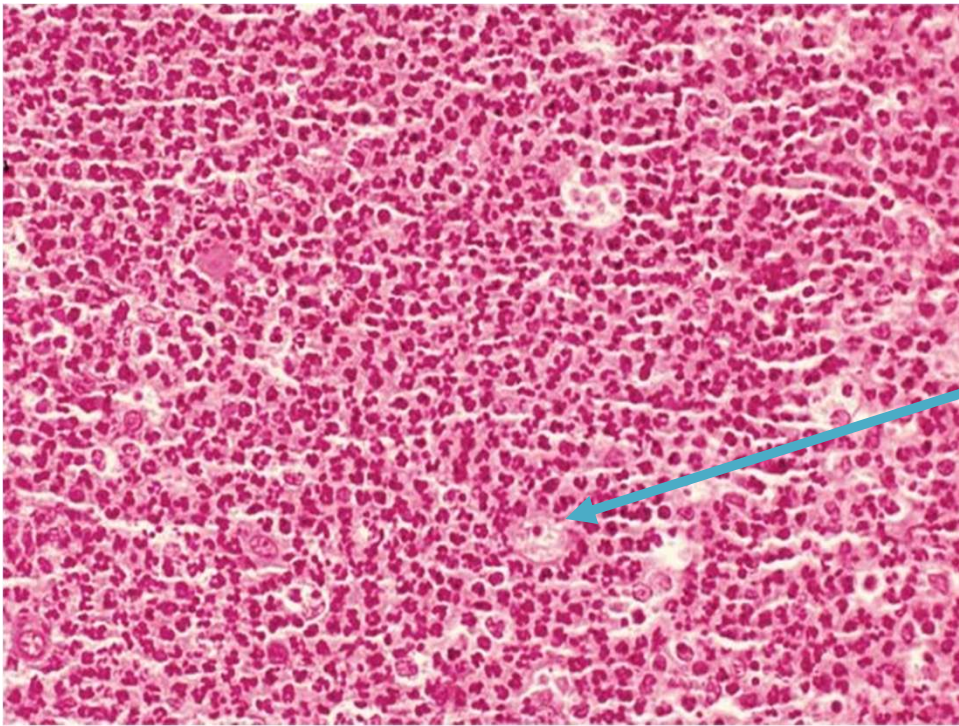
Plasma Cell



Histocyte (Foamy)

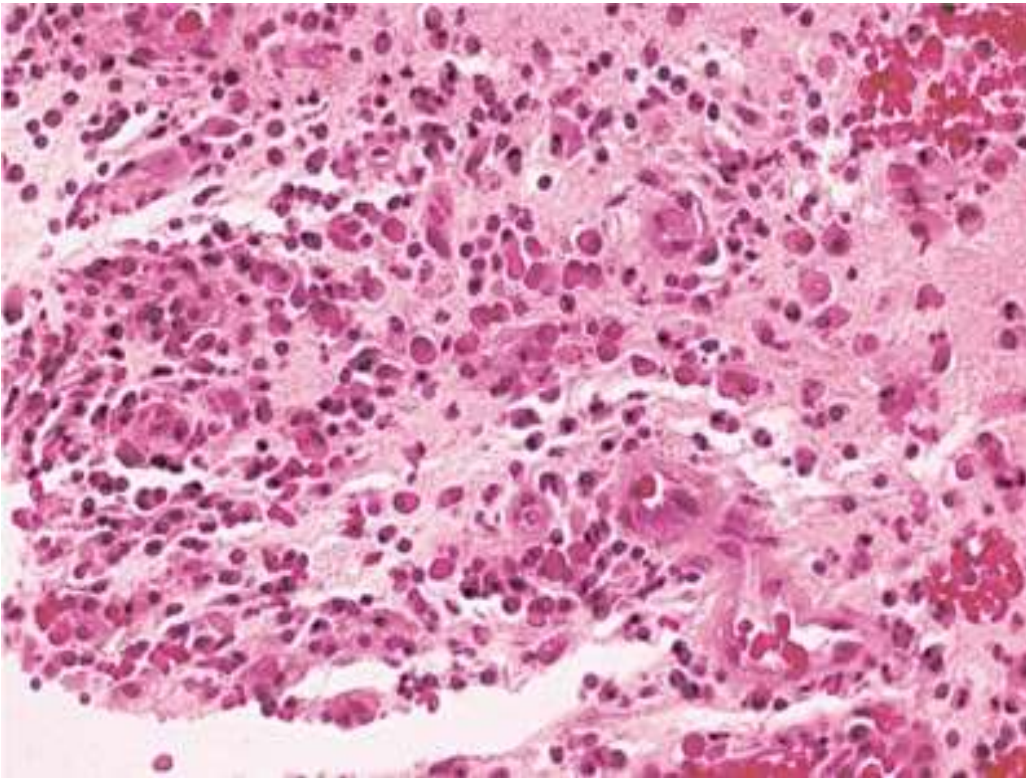


Periapical Abscess



- Mixture of **neutrophils**, cellular debris, bacterial colonies, and necrotic debris.
 - Foamy histocytes can be present in some cases.

Periapical Granuloma



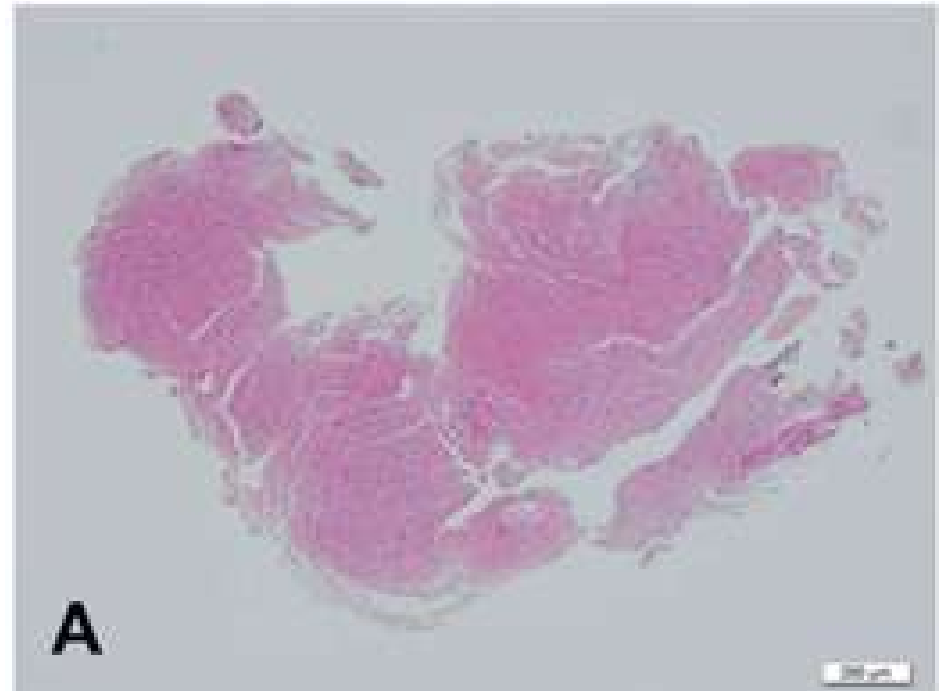
Oral and Maxillofacial Pathology 5th Edition Neville

Histology

- **Granulation tissue containing chronic** or subacute **inflammation**
 - Typical cells identified:
 - **Chronic – lymphocytes and plasma cells**
 - Subacute – neutrophils mixed with **chronic inflammatory cells**
 - Histocytes may also be seen with the granulation tissue

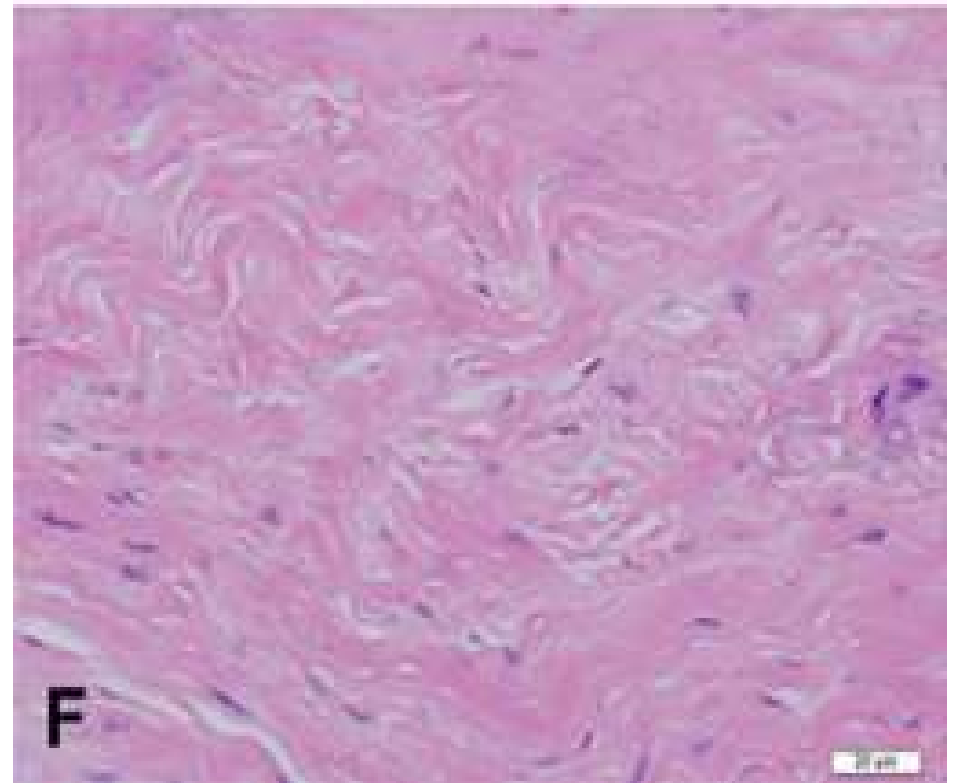
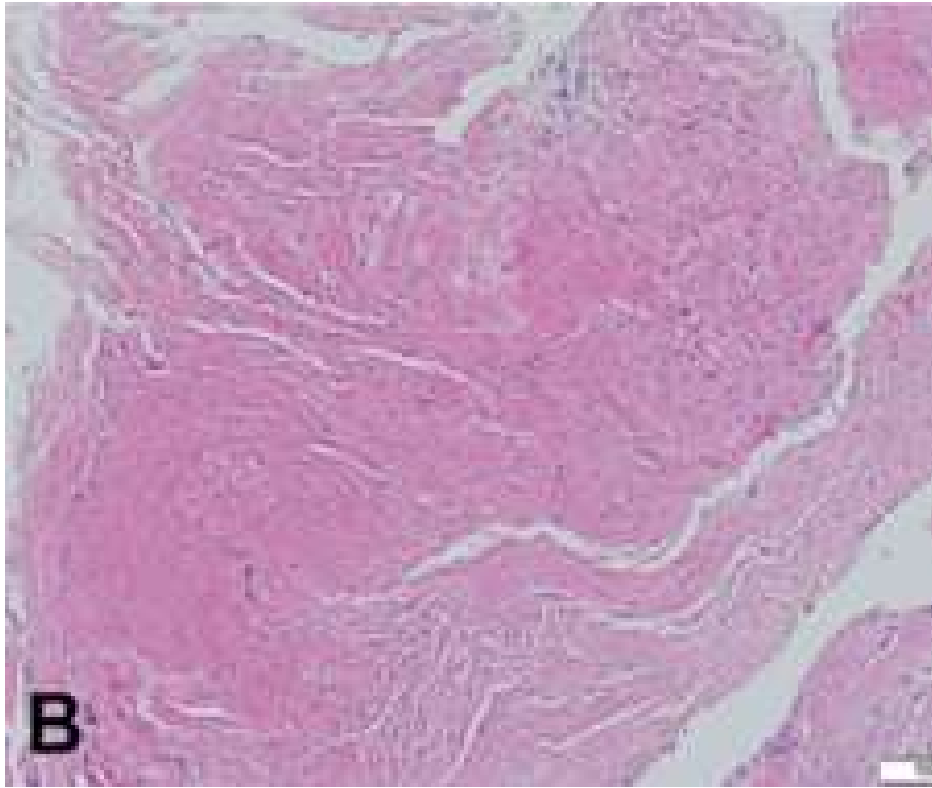
Periapical Scar

- Occurs when **scar tissue replaces normal bone** at the apex of a tooth.
 - Reparative response to periapical inflammation
- Appearance of **hyalinized fibrous connective tissue** (scar)



Journal of Dental Sciences Volume 16, Issue 4, October 2021, Pages 1140-1145

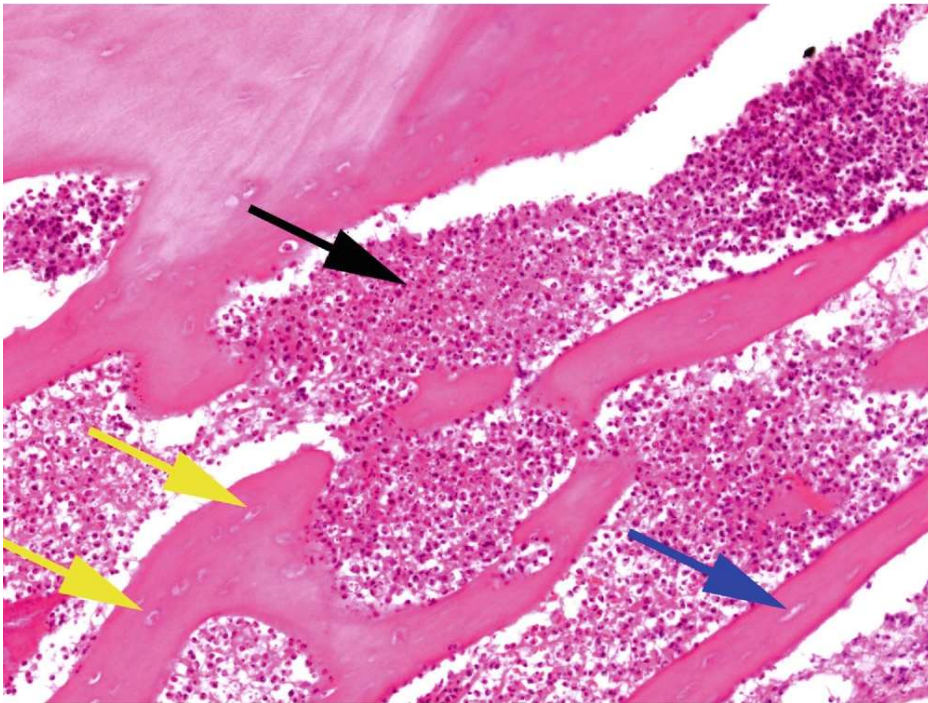
Periapical Scar



Journal of Dental Sciences Volume 16, Issue 4, October 2021, Pages 1140-1145

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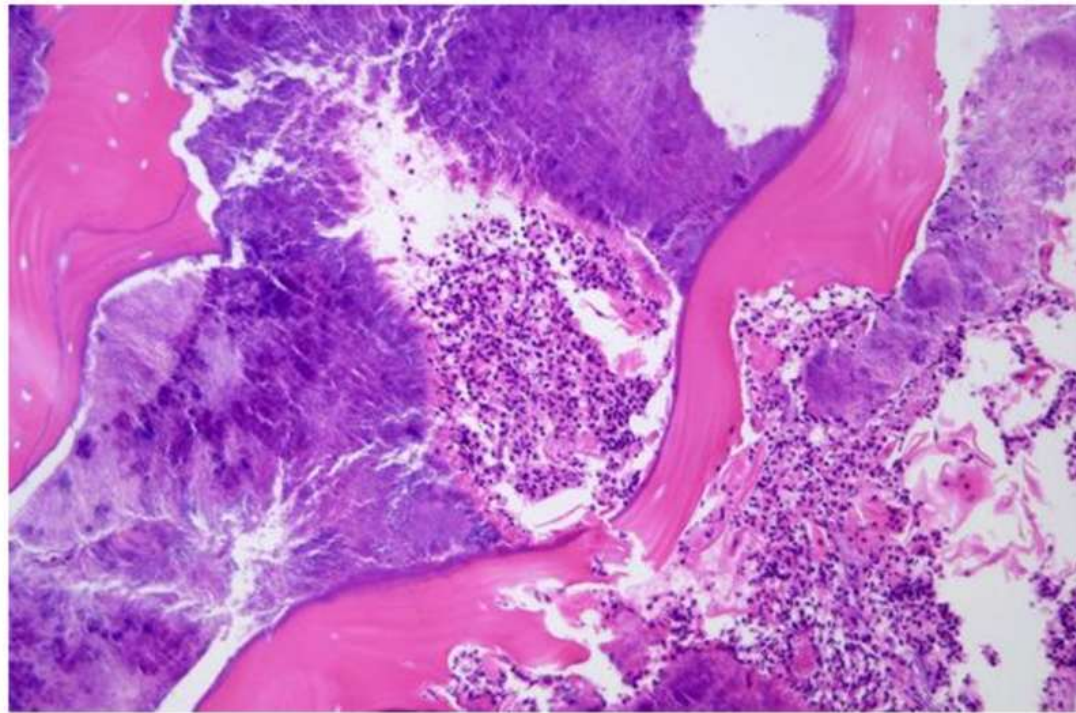
Acute Osteomyelitis



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- Histologic features
 - **Necrotic bone** (yellow arrow) with loss of osteocytes in the lacunae (blue arrow)
 - The **bone has “ragged” borders**
 - The bone is surrounded **acute inflammation composed of neutrophils** (black arrow)
 - **Bacterial colonies** presenting as **purple aggregates** can also be seen.

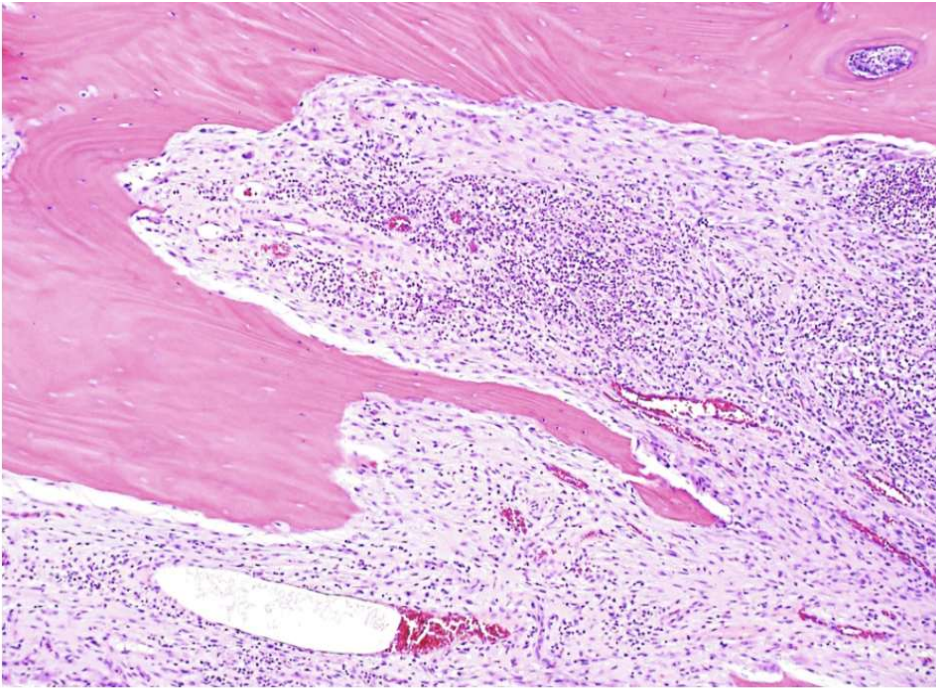
Acute Osteomyelitis with Bacterial Colonies



International Journal of Dentistry January 20102010(2):702314

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Chronic Osteomyelitis



- Granulation tissue found between portions of lamellar bone
 - **Chronic or subacute inflammation** of the granulation tissue

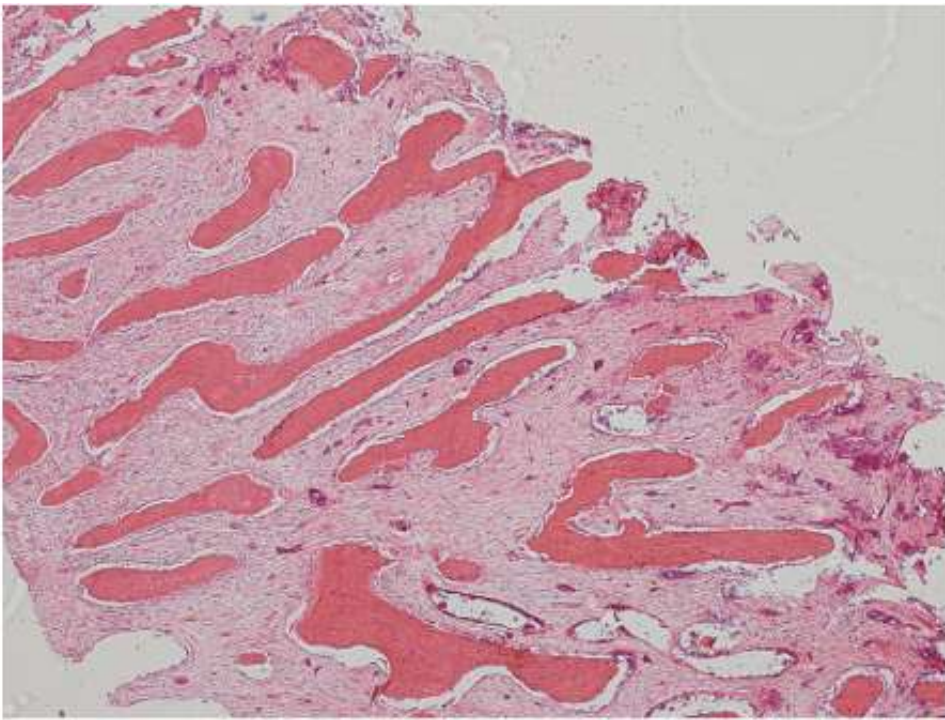
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Chronic Osteomyelitis with Proliferative Periostitis

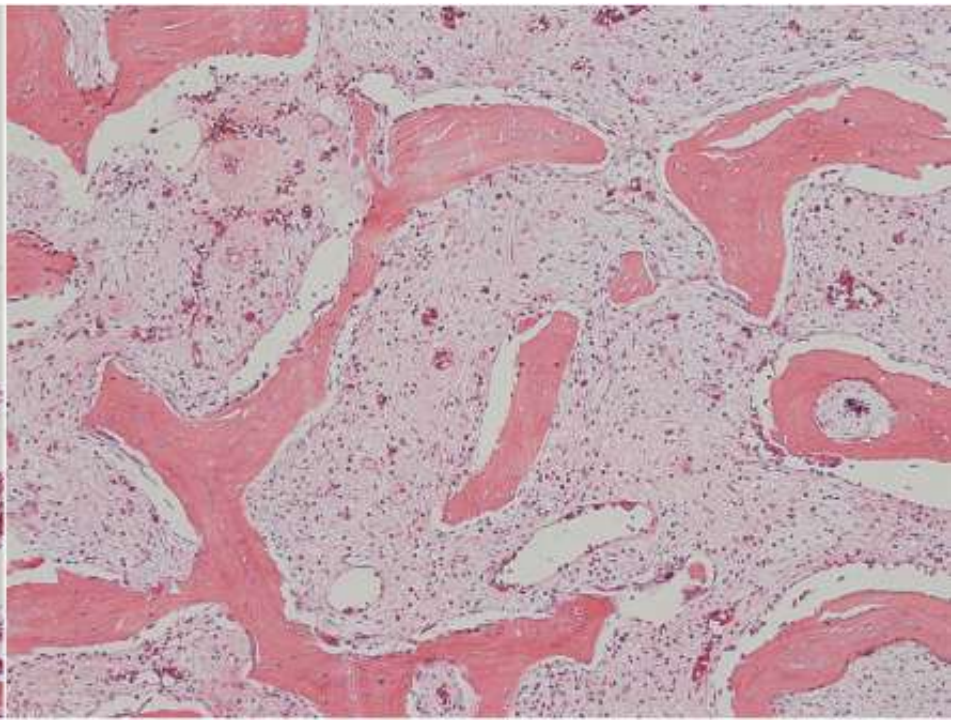
“Onion Skin”

- Proliferative periostitis = Garre’s Osteomyelitis
- Caused by **new bone formation and reaction of the periosteum** certain disease of bone (inflammatory and neoplastic)
- New bone develops parallel to the disease bone giving a layered or “onion skin appearance”.

Chronic Osteomyelitis with Proliferative Periostitis “Onion Skinning”

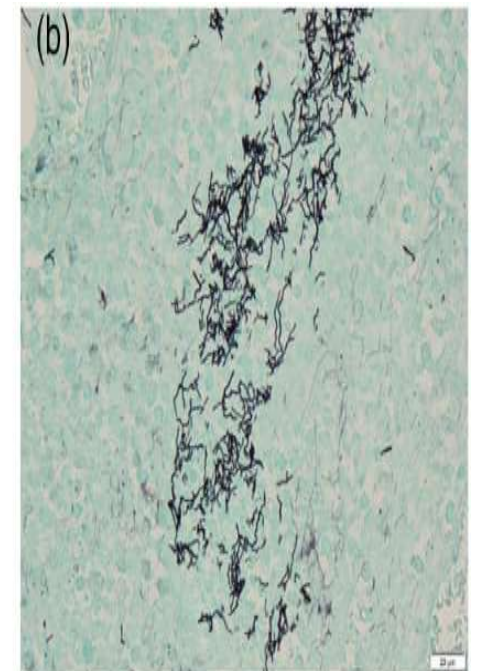
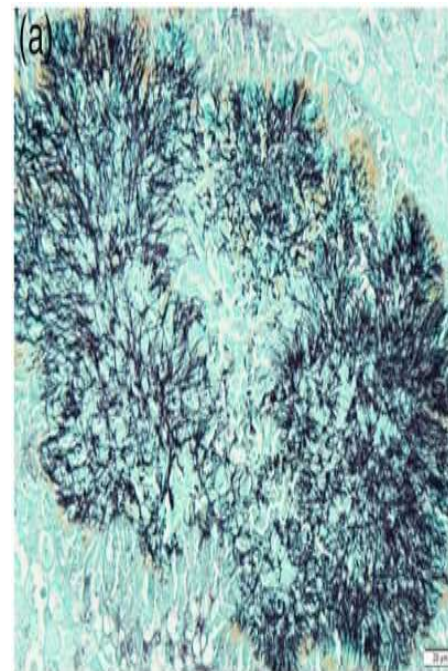
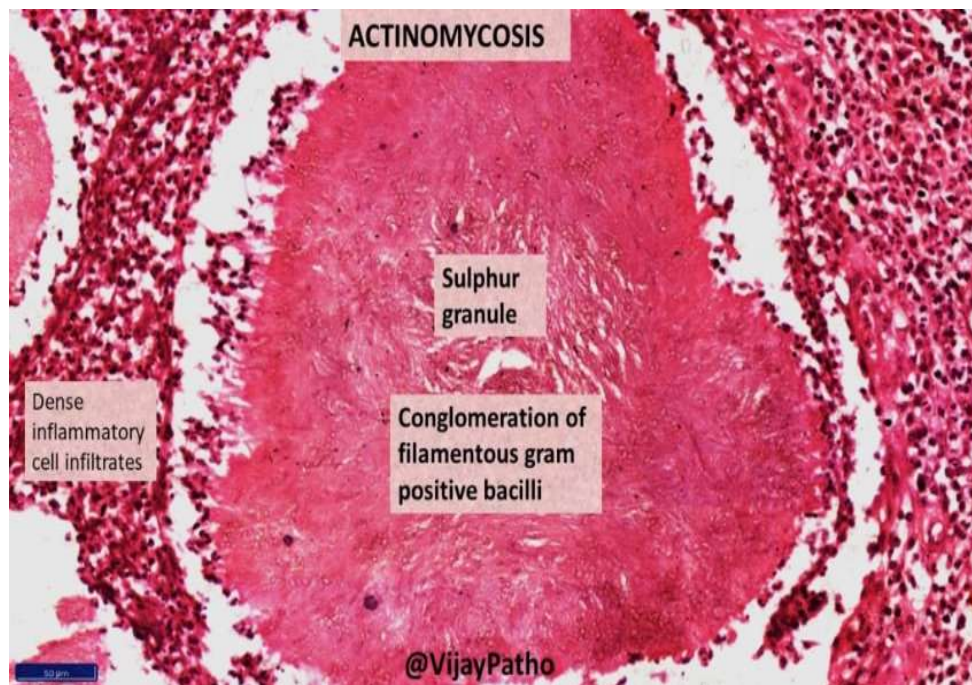


Proliferative periostitis “onion skin”



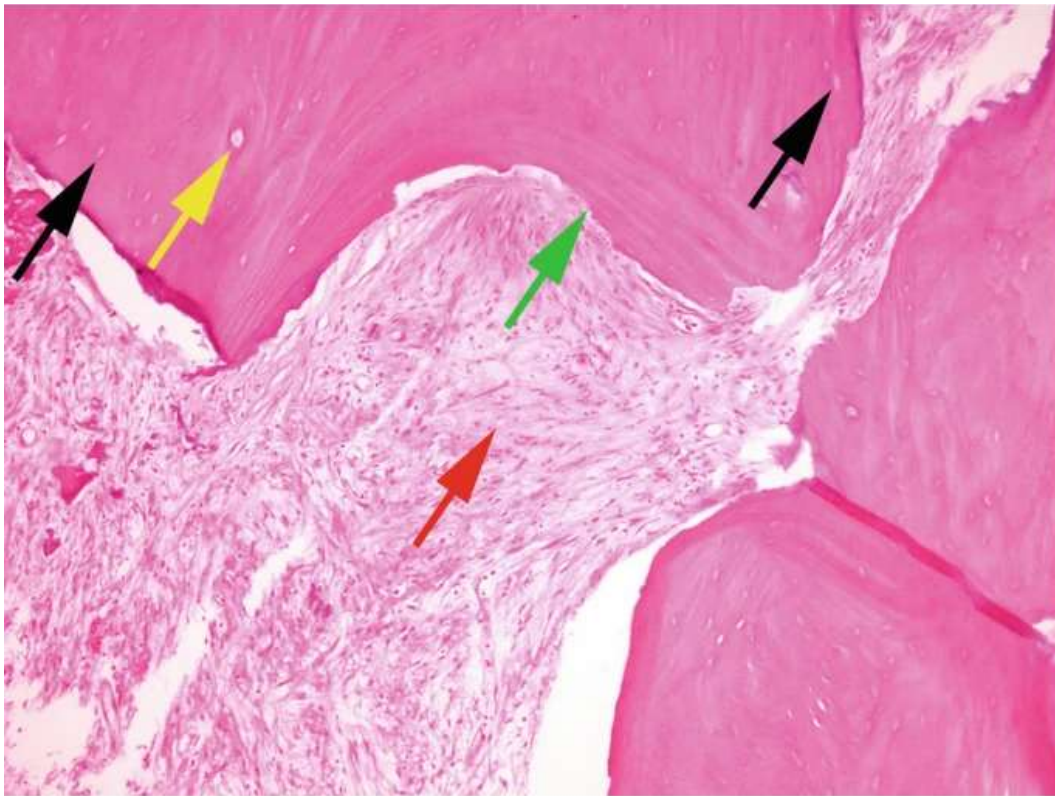
Lamellar bone with osteomyelitis

Osteomyelitis with Actinomycosis



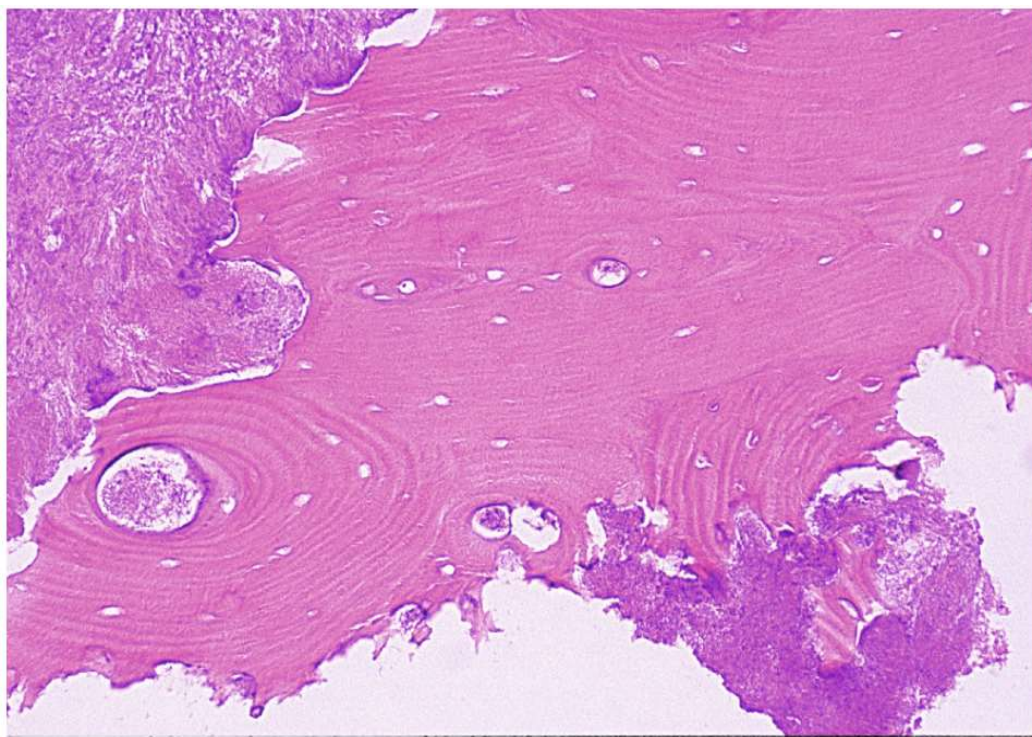
Chew, Siu-Jun et al. "Paediatric actinomycosis: A 16-year, single-institution retrospective review of cases." *Journal of paediatrics and child health* vol. 59,6 (2023): 833-839.

Osteoradionecrosis of the Jaw (ORN)



- Histologic features
 - **Necrotic bone** with loss of osteocytes in the lacunae (black arrow)
 - The **bone has “ragged” borders**
 - Granulation tissue containing **acute inflammation composed of neutrophils** (black arrow)
 - Missing osteoblastic rimming (green arrow)
 - Scar tissue replacing bone marrow (red arrow)

Medication-Related Osteonecrosis of the Jaw (MRONJ)

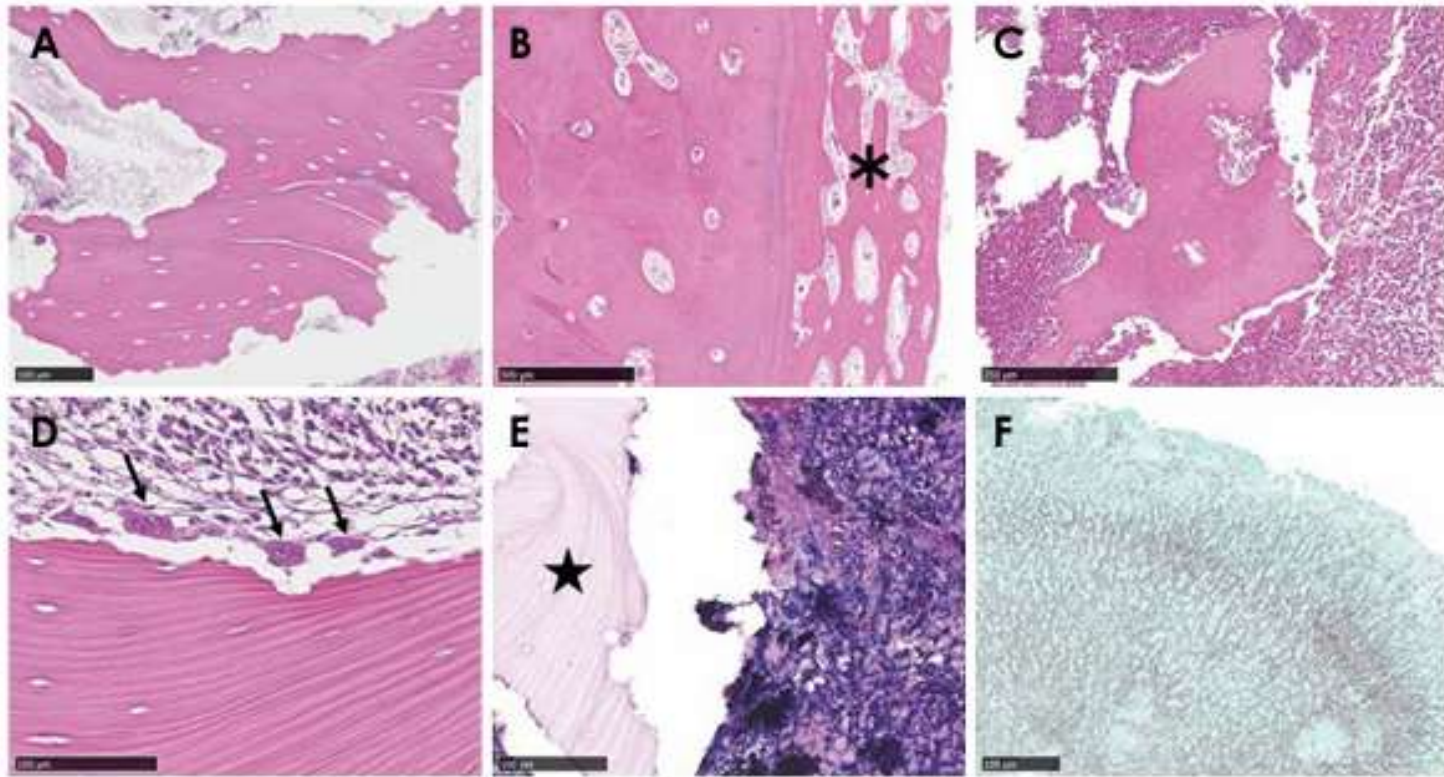


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Histology

- Sclerotic bone with feature of necrosis
 - Necrotic features - loss of osteocytes and “ragged borders”
- Subacute inflammation found in surrounding granulation tissue
- Purple bacterial colonies are present

MRONJ



Ogura, Ichiro et al. Imaging science in dentistry vol. 51,1 (2021): 73-80.
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Resources

- Chapter 3: Pulpal and Periapical Disease
- Chapter 5: Actinomycosis
- Chapter 8: Physical and Chemical Injury (MRON and ORN)

